

Kunal Kaushik

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EDUCATION

University of Illinois Urbana-Champaign
B.S. Electrical Engineering, Minor in Computer Science

Aug 2026 – May 2029

WORK EXPERIENCE

Electrical Engineer, BMS

Aug 2026 – Present

Illini Electric Motorsports

- Proposed isoADC precharge sensing at design review for differential accumulator-to-inverter ΔV monitoring; calculated voltage-divider ratios to identify when inverter voltage reaches within 5% of accumulator voltage for precharge completion
- Designed 600V contactor weld-detection schematic and PCB layout using shared pull-up/down circuitry and voltage dividers; verified all 5 contactor/fuse states in LTspice and against hand calculations
- Designed and fabricated STM32 USB-C-to-CAN development board with 3 TCAN1044 channels and 5V/3.3V rails; resolved shared USB/CAN pin conflicts using isolation circuitry
- Hand-soldered CAN development board and completed initial electrical bring-up through continuity checks and verification of 5V/3.3V power rails

Undergraduate Researcher, NASA CANVAS

Aug 2026 – Present

Novel Mobile Robotics Laboratory under Professor Justin Yim

- Developing Venus traversal robot using distributed propeller thrust to roll between stable faces of a rounded tetrahedron
- Designed isogrid chassis and reconfigurable propeller mounts to minimize mass and preserve airflow during rolling
- Specified ESP32, BNO085 IMU over I2C, and existing H-bridge drivers for low-cost control of all 8 unidirectional motors
- Developed sequential activation strategy for 8 unidirectional propeller motors to control transitions between resting states

Hardware Engineer, Self-Balancing Wheeled Humanoid

Jan 2025 – Jul 2025

Nova Robotics LLC

- Prototyped reaction-wheel humanoid hardware; demonstrated balance on a single-wheel inverted-pendulum prototype
- Integrated Raspberry Pi, ODrive S1 drivers, and dual reaction wheels over CAN; built a 6:1 gearbox and custom quadrature-encoder mount for mechanical alignment alongside MPU9050 IMU feedback in the robot balance architecture
- Diagnosed an incorrectly adjusted 24V buck converter with a multimeter; identified unsafe Raspberry Pi supply voltage
- Built modular robot electronics using soldered perfboard and custom wiring harnesses; routed power, sensor, CAN, and I2C connections through constrained electromechanical packaging

PROJECTS

Designer, Open Source BLDC Motor Controller

Apr 2026 – Present

- Architected ESP32-based BLDC controller schematic with three-phase gate driving and high-resolution encoder feedback
- Designed DRV8323SRTAR and CSD MOSFET power stage for a target 12V–60V supply range; specified buck/LDO logic rails, bulk capacitance, ESD protection, and reverse-polarity protection
- Created custom PCB footprints for components unavailable in EDA libraries to support future controller board layout

Project Lead, Competition Robot

Jun 2024 – Nov 2025

- Architected and assembled 10-DoF robot integrating sensing, vision, and motor subsystems for autonomous manipulation
- Designed sprung cable routing for linear slides to maintain repeatable wire position through full mechanism extension; routed servo power through a dedicated module to address voltage droop during simultaneous robot subsystem operation
- Restored failed control hub by isolating a poorly soldered battery connector with a multimeter and resoldering the joint

Spatial Snakes | RoMeLa at UCLA

Jun 2026 – Aug 2026

- Built Spatial Snakes prototype under Professor Dennis Hong using MIT CSAIL's y-zipper as the extrusion mechanism
- Designed ESP32-actuated slider and fabricated custom breakout PCB for static actuation and peripheral control; programmed concurrent actuator monitoring with error detection logic to prevent fabric sheath snags and tears
- Created everting fabric sheath spooled by stepper motor to grow from the extension tip and reduce footprint by 30%

SKILLS

EDA/CAD & Simulation: Altium, KiCad, LTspice, EasyEDA, Fusion 360, Onshape

Embedded & Programming: C/C++, Python, MATLAB, STM32, ESP32, Raspberry Pi

Protocols & Hardware: Schematic Design, PCB Layout, CAN, I2C, SPI, UART, PWM, ADC

Test & Build: Oscilloscope, Multimeter, Bench Supply, Soldering, SMT/Reflow, Crimping, FDM Printing, CNC, DFM/DFA